Human Practice Sharing session

OUR ORCA and FREQUENCY-SPECTURM INCLUSIVITY

SUSTech-BIO
HP Summit
30, August, 2026

Where does our ORCA come from?

How we design the project name and mascot

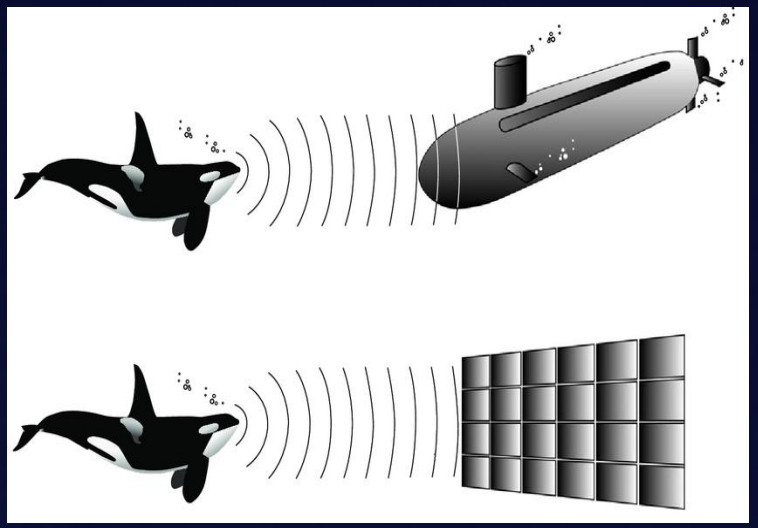


On-target
RNA
Correction of
APOE4
for Alzheimer's Disease



ORCA, the killer whales!

- Matriarchal pods — led by elder females
- Grandmothers carry survival memory for the group
- No member is left behind



- Emit sound pulses into the unknown
- Read the echoes that bounce back
- Adjust course based on what they hear

Acceptor or Giver?

Apr 16 · Shenzhen Senior Care Center

A mini class on scam prevention & intro to Biology for the elderly





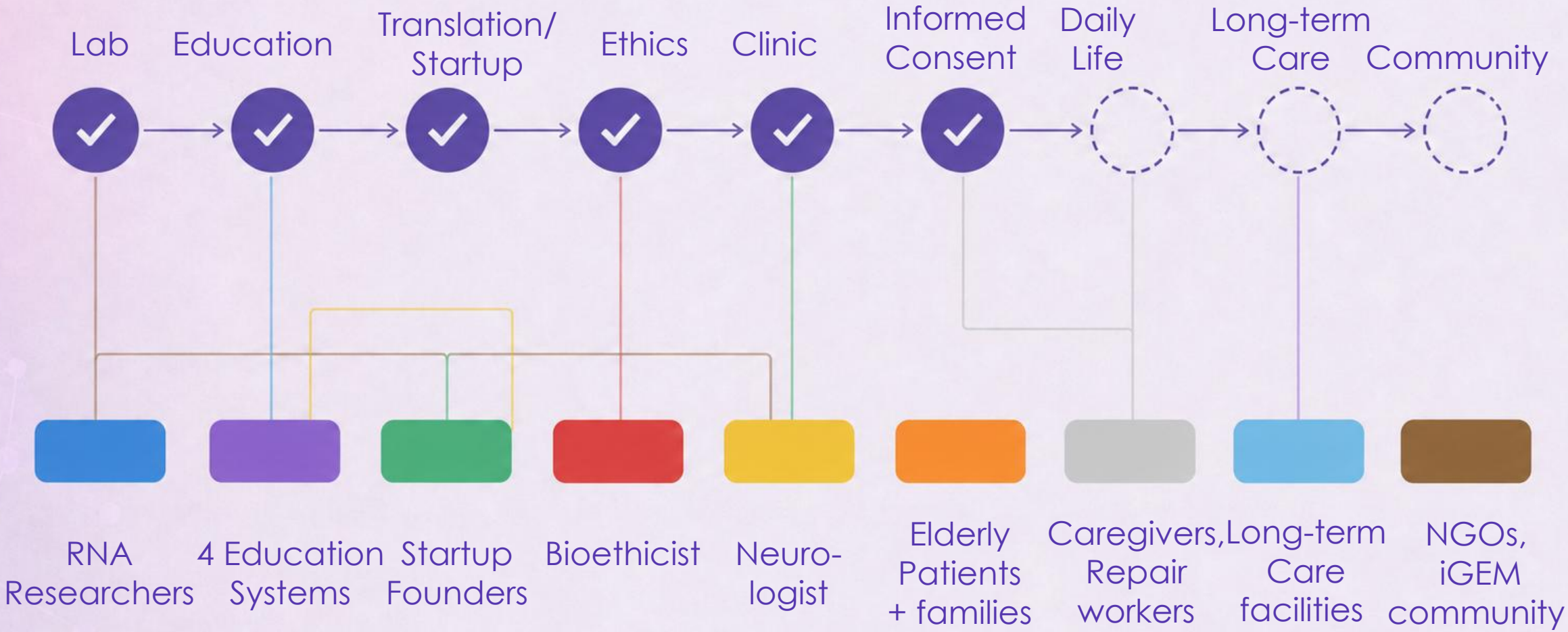


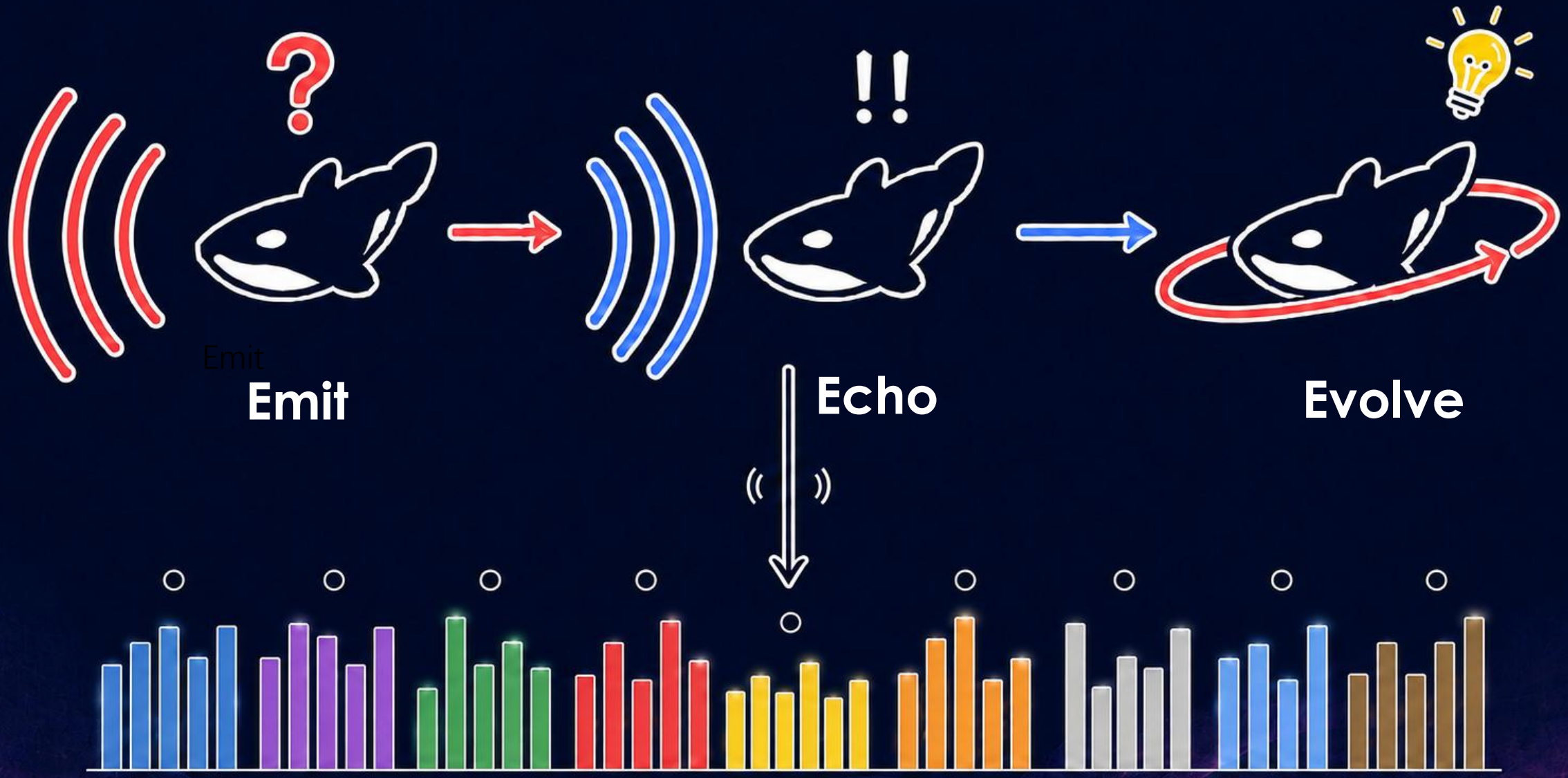
“So… what is this for? "

— An elderly man, after politely completing our DNA extraction (from bananas) activity

Stakeholder Map

From Lab to Life





Framework Overview of HP and Inclusivity

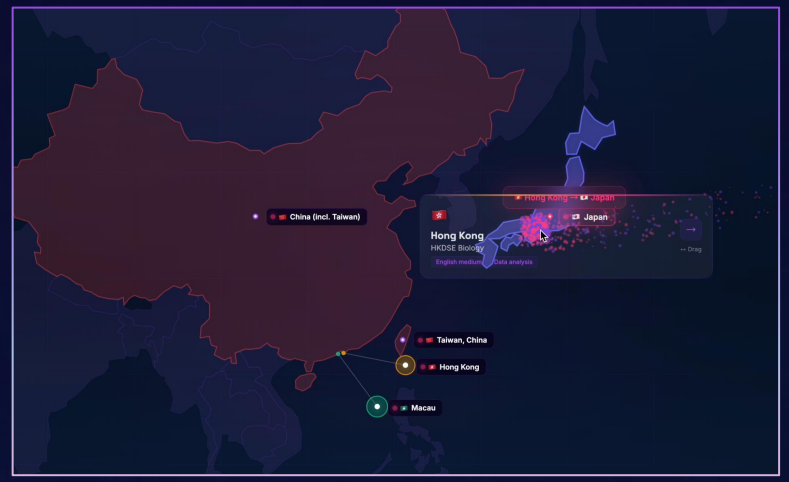
EEE & Frequency Spectrum

Cycle 1: Who gets to understand?

EEE & Frequency Spectrum

4 Education Systems Study

CUHK-Hongkong-SBS × HG-Tokyo × SUSTech-BIO



Few Bioethics Coverage

- Gaokao
- HKDSE
- Macao
- Tokyo
- 100+ docs
- 4 systems
- 15+ reviews
- cross-transplant experiments

Public Cognitive Survey (In Progress)

3 dimensions across age groups & education backgrounds:

- Know — Have you heard of RNA editing?
- Explain — Can you explain what it does?
- Decide — Could you evaluate a treatment decision?

Target: cross-age, cross-education sample

Cycle 2: Listen to the Full Chain

EEE & Frequency Spectrum

Role	Quotes	Changed Actions
Startup CEO	"Patients sign consent form they don't understand"	Drove escape room game design
Neurologist	"Early intervention >> late treatment"	Reframed to "risk intervention"
Ethicist	"Telling someone they carry APOE4 is itself an intervention"	Added residual risk disclosures
Repair Team Leader	"Not knowledge — physical capacity"	Fed aging-friendly app logic

Failed： Collaboration with Shokz

Cycle 3: Stress-Testing in the Real World

WORLDPROOF BIOHackathon × iGEM Village

Build it. Break it. Fix it.







- 50%+ / non-bio at WORLDPROOF
- 3.2 / avg. assumptions revised per team
- Non-bio participants could be better Red Teamers



1st Grand APiC

- 600+ participants ·SUSTech-BIO as co-organizer

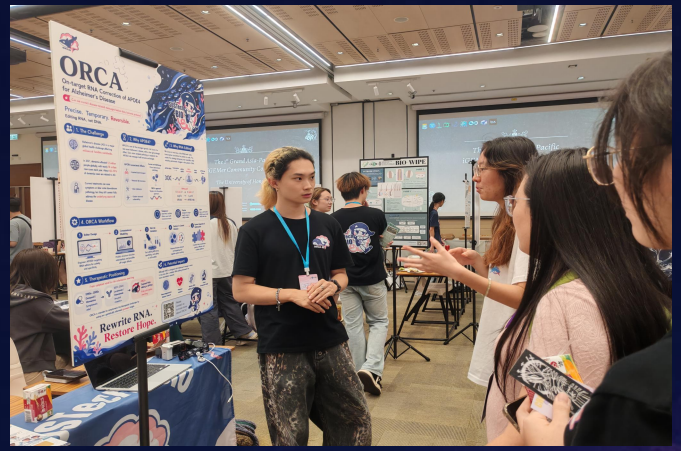
Echoes from the Community







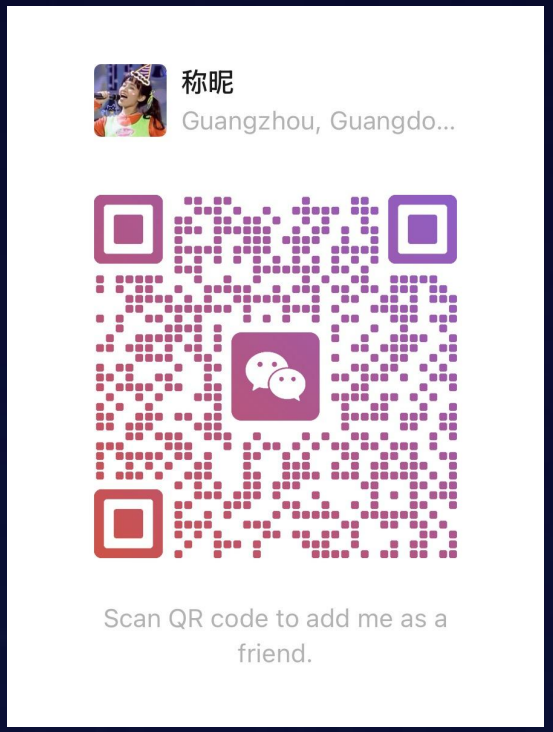






What we can build together

- **Aging-Friendly Design** — We heard from a women's repair team that elderly people know the hazards but can't fix them. We have the insight, not the solution. Co-design with us.
- **Multi-Version Informed Consent** — Plain text, audio, visual, proxy decision-maker. We're prototyping. Any biotech project can plug in and contribute.
- **Escape Room for SynBio Literacy** — A game that makes people experience the consent problem, not just hear about it. Theme is ours, format is yours.
- Joint Cognition Survey — We're building a public survey on RNA editing awareness. Add your topic, share respondents, everyone gets a bigger dataset.
- **HP Red Team** — You break our HP logic, we break yours. Both walk into Jamboree sharper. Uses WORLDPROOF rules.



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Build Together!

Thank You!

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